



INTRODUCTION TO DS, ML AND AI

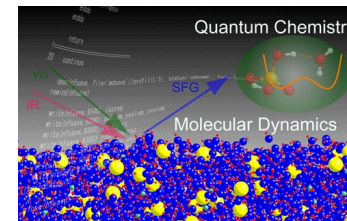
Prof. Raghunathan Rengaswamy
Indian Institute of Technology Madras

Big Data

- Large amounts of data
- Size of data is so huge that traditional data management tools are unable to store, manage and process the data
- Pillar behind the idea that one can make useful inference with a large body of data that wasn't possible before with smaller datasets
- Extremely large datasets can be analyzed computationally to reveal patterns, trends, and associations that are not transparent or easy to identify otherwise

Big Data is Everywhere!

- Enormous amounts of data are being collected and stored
 - Web data, e-commerce
 - Point-of-sale at stores
 - Bank transactions
 - Social Network
 - Real-time plant data
 - Materials Formulation & Design
 - Molecular Simulations



Characteristics of Big Data- IBM 4 Vs



Volume



Velocity



Variety



Veracity



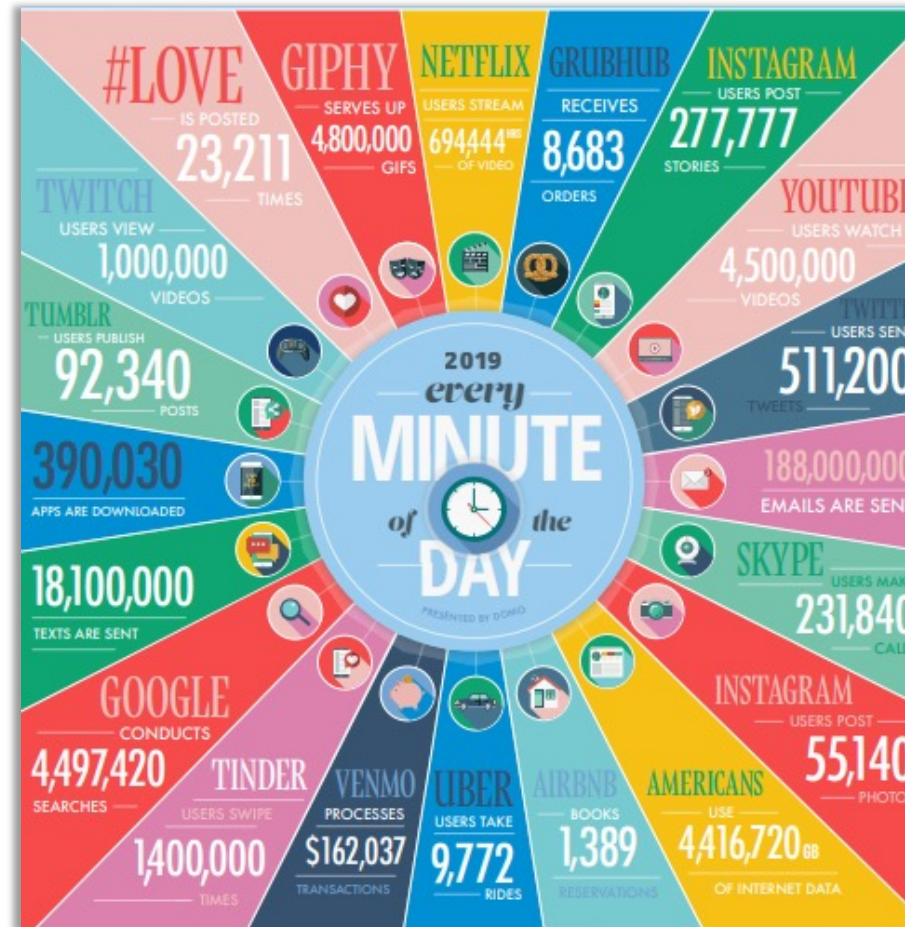
Big Data – Volume

- Currently 2.5 EB (Exabytes) of data is produced everyday
- By 2025, 463 EB of data will be produced everyday
- 90% of the data today was created from the last two years

1000 kB kilobyte	1000 ⁵ PB petabyte
1000 ² MB megabyte	1000 ⁶ EB exabyte
1000 ³ GB gigabyte	1000 ⁷ ZB zettabyte
1000 ⁴ TB terabyte	1000 ⁸ YB yottabyte

Big Data - Velocity

What happens in an internet minute ?



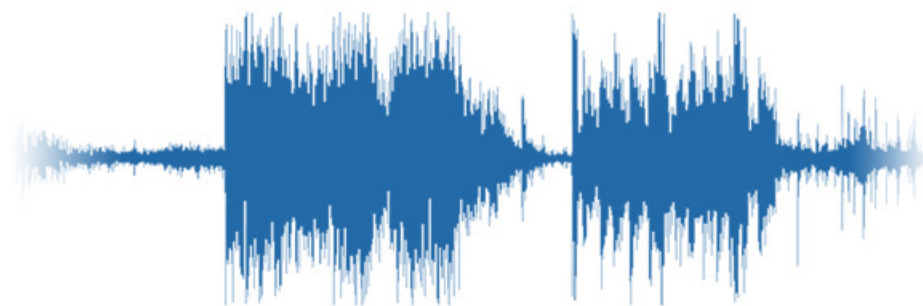
<https://www.domo.com/learn/data-never-sleeps-7>

Big Data - Variety

- Type of data generated is different- structured, semi-structured, unstructured
- In the past, primary sources of data collection were from databases and spread sheet formats like excel, now the data are coming in the form of images, audios, videos, sensors etc.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	age	JobType	EdType	maritalstatus	occupation	relationship	race	gender	capitalgain	capitalloss	hoursperweek	SalStat		
2	19	State-gov	Bachelors	Never-married	Adm-clerical	Husband	White	Male	2174	0	50	less than or equal to 50,000		
3	50	Self-emp-inc	Bachelors	Married-civ-spouse	Exec-managerial	Other-relative	Amer-Indian-Eskimo	Male	0	0	13	less than or equal to 50,000		
4	18	Private	HS-grad	Divorced	Handlers-cleaners	Ovns-child	Black	Male	0	0	40	less than or equal to 50,000		
5	53	Private	11th	Married-civ-spouse	Handlers-cleaners	Not-in-family	Black	Male	0	0	40	less than or equal to 50,000		
6	28	Private	Bachelors	Married-civ-spouse	Prof-specialty	Wife	Asian-Pac-Islander	Female	0	0	40	less than or equal to 50,000		
7	37	Private	Masters	Married-civ-spouse	Exec-managerial	Unmarried	Amer-Indian-Eskimo	Female	0	0	40	less than or equal to 50,000		
8	49	Private	9th	Married-spouse-absent	Other-service	Unmarried	Asian-Pac-Islander	Female	0	0	16	less than or equal to 50,000		
9	52	Self-emp-inc	HS-grad	Married-civ-spouse	Exec-managerial	Other-relative	Black	Male	0	0	45	greater than 50,000		
10	31	Private	Masters	Never-married	Prof-specialty	Husband	White	Female	14084	0	50	greater than 50,000		
11	42	Private	Bachelors	Married-civ-spouse	Exec-managerial	Husband	Asian-Pac-Islander	Male	5178	0	40	greater than 50,000		
12	37	Private	Some-colh	Married-civ-spouse	Exec-managerial	Wife	Black	Male	0	0	40	greater than 50,000		
13	30	State-gov	Bachelors	Married-civ-spouse	Prof-specialty	Ovns-child	White	Male	0	0	40	greater than 50,000		
14	23	Private	Bachelors	Never-married	Adm-clerical	Not-in-family	Amer-Indian-Eskimo	Female	0	0	30	less than or equal to 50,000		
15	32	Private	Assoc-acd	Never-married	Sales	Unmarried	Black	Male	0	0	50	less than or equal to 50,000		
16	34	Private	7th-9th	Married-civ-spouse	Transport-moving	Other-relative	Asian-Pac-Islander	Male	0	0	45	less than or equal to 50,000		
17	25	Self-emp-inc	HS-grad	Never-married	Farming-fishing	Husband	Black	Male	0	0	35	less than or equal to 50,000		
18	32	Private	HS-grad	Never-married	Machine-op-inspct	Husband	Asian-Pac-Islander	Male	0	0	40	less than or equal to 50,000		
19	38	Private	11th	Married-civ-spouse	Sales	Unmarried	Other	Male	0	0	50	less than or equal to 50,000		
20	43	Self-emp-inc	Masters	Divorced	Exec-managerial	Other-relative	Other	Female	0	0	45	greater than 50,000		
21	40	Private	Doctorate	Married-civ-spouse	Prof-specialty	Husband	White	Male	0	0	60	greater than 50,000		
22	54	Private	HS-grad	Separated	Other-service	Wife	Other	Female	0	0	20	less than or equal to 50,000		
23	35	Federal-gov	9th	Married-civ-spouse	Farming-fishing	Other-relative	White	Male	0	0	40	less than or equal to 50,000		
24	43	Private	11th	Married-civ-spouse	Transport-moving	Ovns-child	White	Male	0	2042	40	less than or equal to 50,000		
25	59	Private	HS-grad	Divorced	Tech-support	Unmarried	Amer-Indian-Eskimo	Female	0	0	40	less than or equal to 50,000		
26	56	Local-gov	Bachelors	Married-civ-spouse	Tech-support	Unmarried	Black	Male	0	0	40	greater than 50,000		
27	19	Private	HS-grad	Never-married	Craft-repair	Ovns-child	Black	Male	0	0	40	less than or equal to 50,000		
28	54	?	Some-colh	Married-civ-spouse	?	Not-in-family	Other	Male	0	0	60	greater than 50,000		
29	39	Private	HS-grad	Divorced	Exec-managerial	Other-relative	White	Male	0	0	80	less than or equal to 50,000		
30	49	Private	HS-grad	Married-civ-spouse	Craft-repair	Wife	Amer-Indian-Eskimo	Male	0	0	80	less than or equal to 50,000		
31	23	Local-gov	Assoc-acd	Never-married	Protective-serv	Unmarried	Other	Male	0	0	52	less than or equal to 50,000		
32	20	Private	Some-colh	Never-married	Sales	Husband	Asian-Pac-Islander	Male	0	0	44	less than or equal to 50,000		
33	45	Private	Bachelors	Divorced	Exec-managerial	Wife	Amer-Indian-Eskimo	Male	0	1408	40	less than or equal to 50,000		
34	30	Federal-gov	Some-colh	Married-civ-spouse	Adm-clerical	Other-relative	White	Male	0	0	40	less than or equal to 50,000		
35	22	State-gov	Some-colh	Married-civ-spouse	Other-service	Other	Other	Male	0	0	15	less than or equal to 50,000		
36	48	Private	11th	Never-married	Machine-op-inspct	Other-relative	Asian-Pac-Islander	Male	0	0	40	less than or equal to 50,000		
37	21	Private	Some-colh	Never-married	Machine-op-inspct	Ovns-child	Asian-Pac-Islander	Male	0	0	50	less than or equal to 50,000		
38	19	Private	HS-grad	Married-AF-spouse	Adm-clerical	Ovns-child	Amer-Indian-Eskimo	Female	0	0	25	less than or equal to 50,000		
39	48	Self-emp-inc	Assoc-acd	Married-civ-spouse	Prof-specialty	Unmarried	Asian-Pac-Islander	Male	0	0	40	less than or equal to 50,000		

Excel



Audio file (mp3)

Big Data - Veracity

- Uncertainty in data due to inherent inconsistencies
- Usually results from huge volume
- Impossible to develop a central veracity-checking mechanism
- Cases from history
 - Elections where fake data can play a role
 - How many posts in quora represent real experiences?

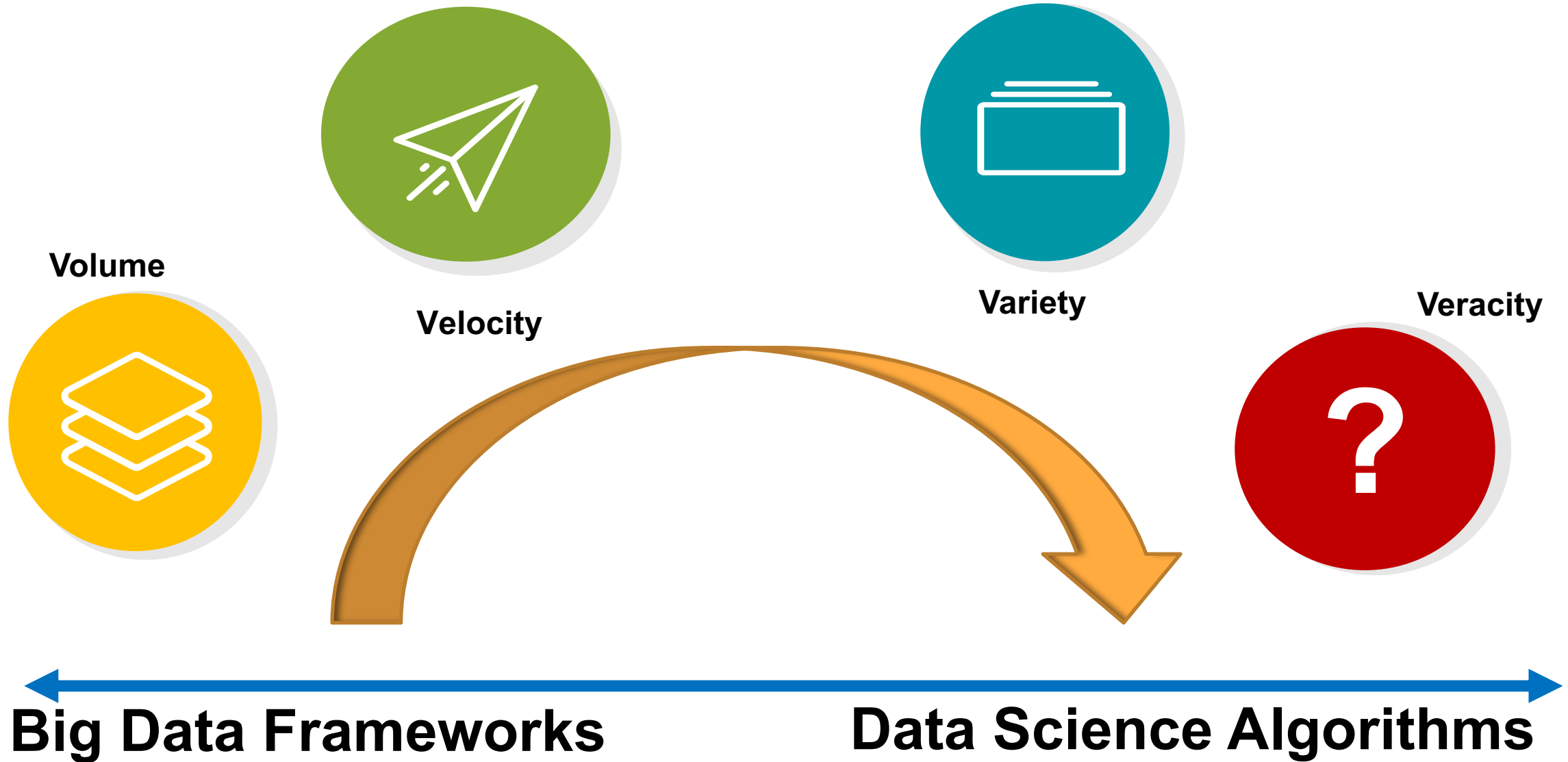
“A lie gets halfway around the world before the truth has a chance to get its pants on” – Winston Churchill

Big Data - Validity (the 5th V) ?

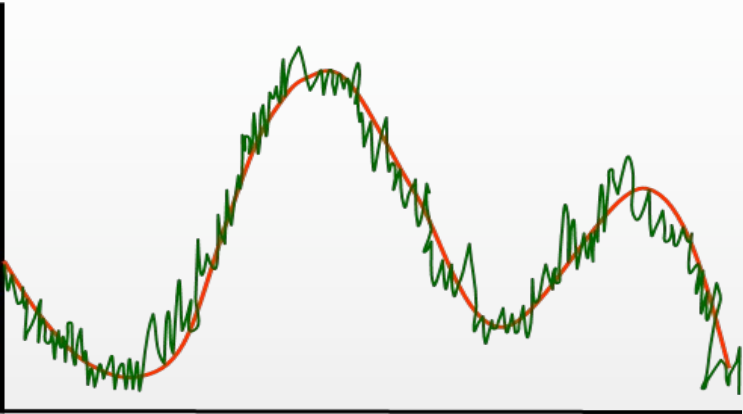
- Google Flu Trends

- Started with the aim to predict flu outbreaks in USA
- Idea was to combine google search queries with the flu tracking information from Centre for Disease Control and Prevention to produce accurate results of flu outbreak
- Failed to predict the outbreak of H1N1 in 2009 and other outbreaks from 2011 onwards
- Searches like 'cold' and 'fever' need not be flu related
- Not all factors responsible for a flu outbreak were considered

“Say not, “I have found the truth,” but rather, “I have found a truth.” – Kahlil Gibran



AI IS FOR EVERYONE



Quantitative Sensor Data



Natural Language
Description –
Oscillating with
decreasing
amplitude,
Increasing linearly,
exponentially



Pergamon

Engng Applic. Artif. Intell. Vol. 8, No. 1, pp. 35–51, 1995
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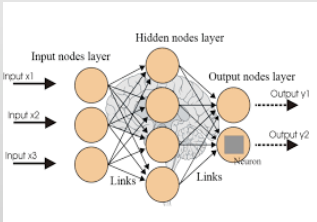
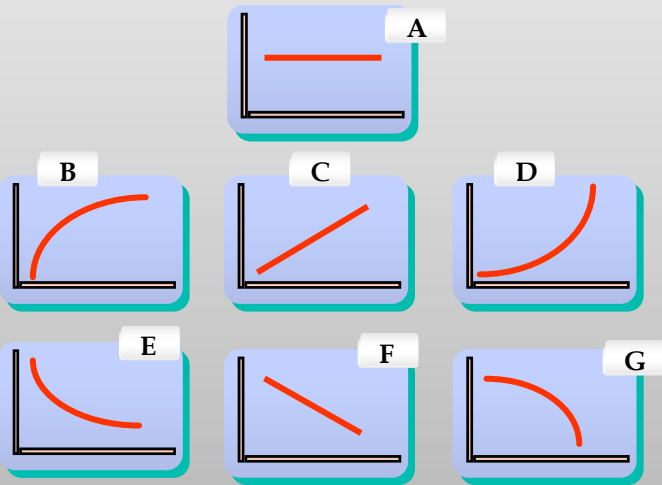
Contributed Paper

A Syntactic Pattern-recognition Approach for Process Monitoring and Fault Diagnosis

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VENKAT VENKATASUBRAMANIAN
Purdue University, West Lafayette, U.S.A.

(Received May 1994; in revised form September 1994)



Context-free and context-
dependent error correction

String to Text

↓
↓
↓
DBGEDBGEDBGEDBAE → Temperature is Oscillating

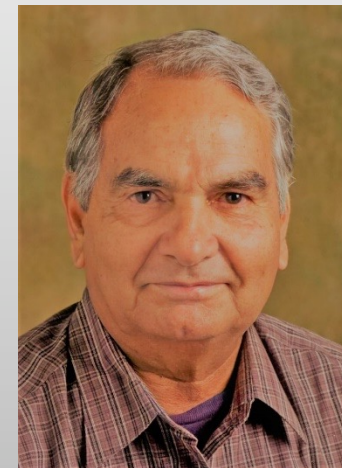
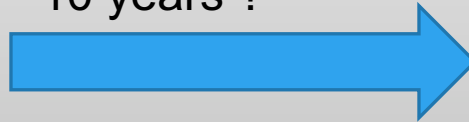
AI IS EVERYWHERE



10 years and half a
Deanship later



What AI thinks will
happen to me in
10 years ?



BROAD DEFINITIONS



What is statistics ?

- The science of working with uncertain data



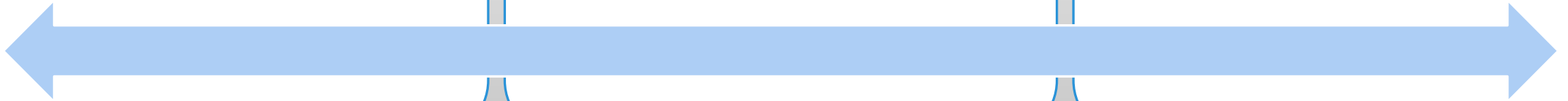
What is data science ?

- The science of deriving insights from data (modelling, predictions)



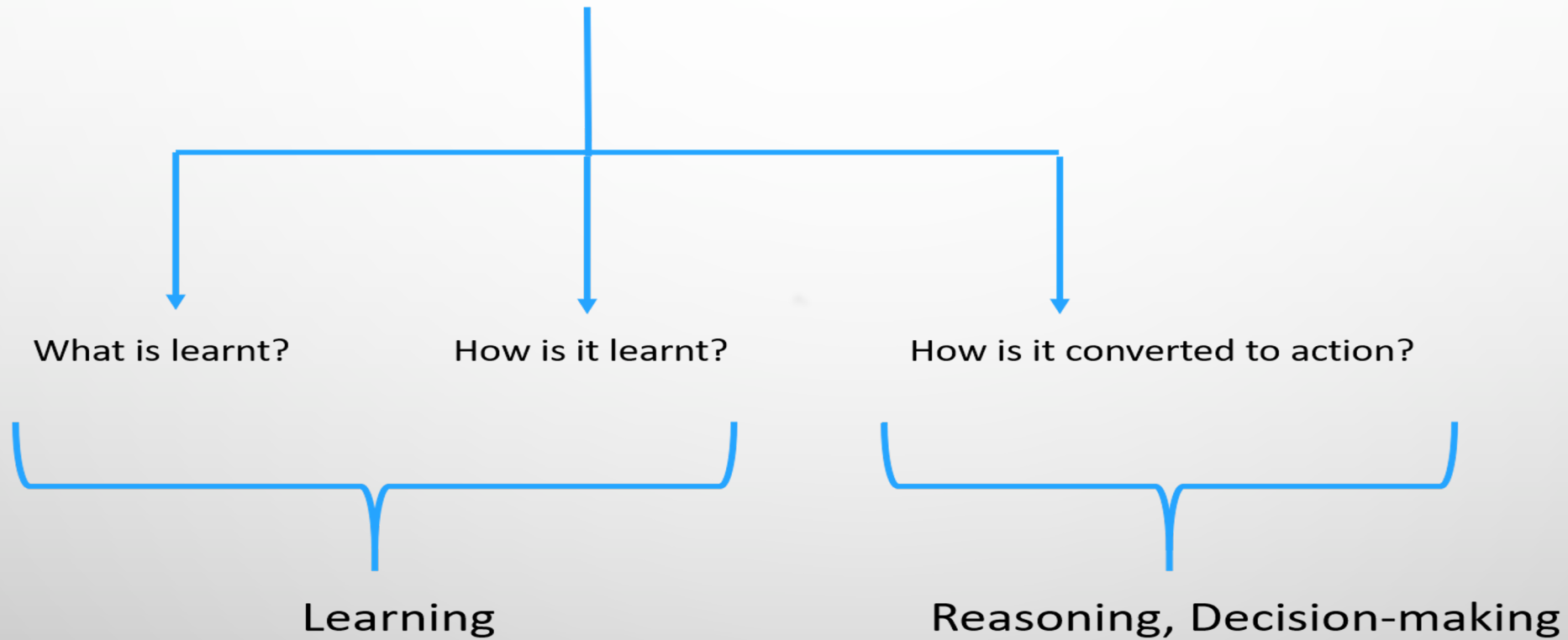
What is ML ?

- The science of mimicking human learning



WHAT IS LEARNING ?

Machine Learning

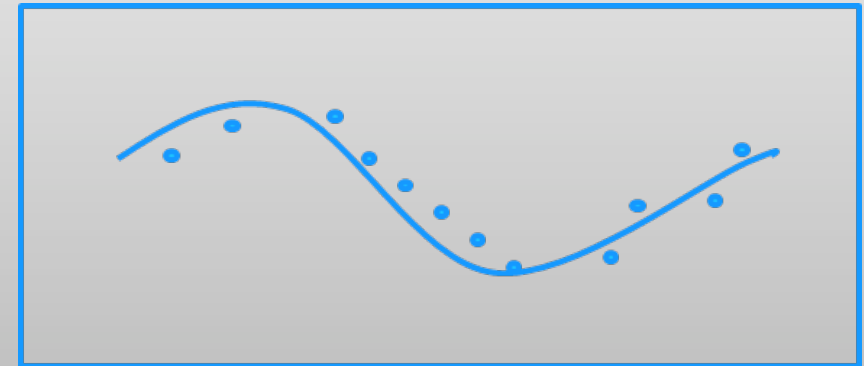
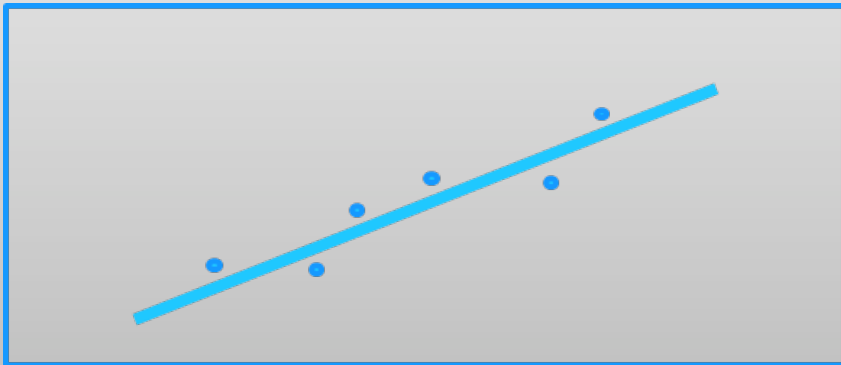


What major problems does ML solve for decision-making ?

Classification problems



Function approximation or Regression problems



MACHINE LEARNING FUNDAMENTALS

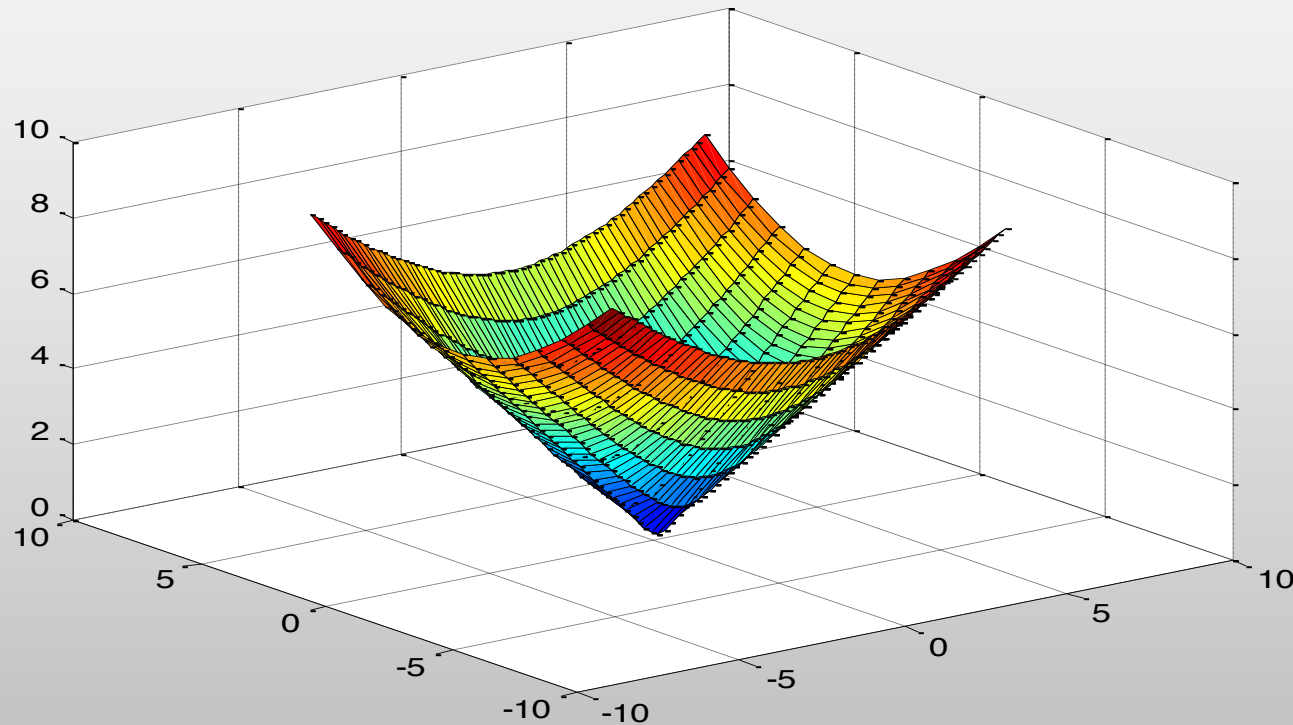
- ML IS ABOUT IDENTIFYING FUNCTION APPROXIMATORS AND DECISION FUNCTIONS UNDER UNCERTAINTY
 - LEADS TO ILL-POSED PROBLEMS
 - TRADE-OFFS ARE REQUIRED

$$A = T + U$$

MIMICKING HOW WE LEARN

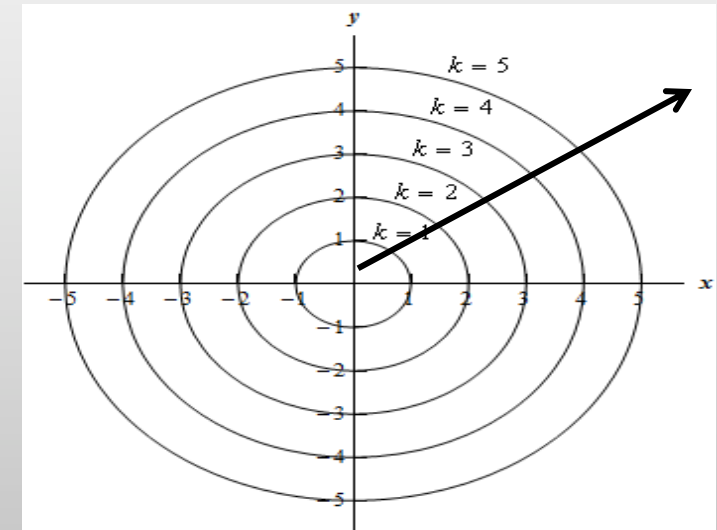
Trial-and-error to choose a function

Optimization to identify the parameters



Contour plot

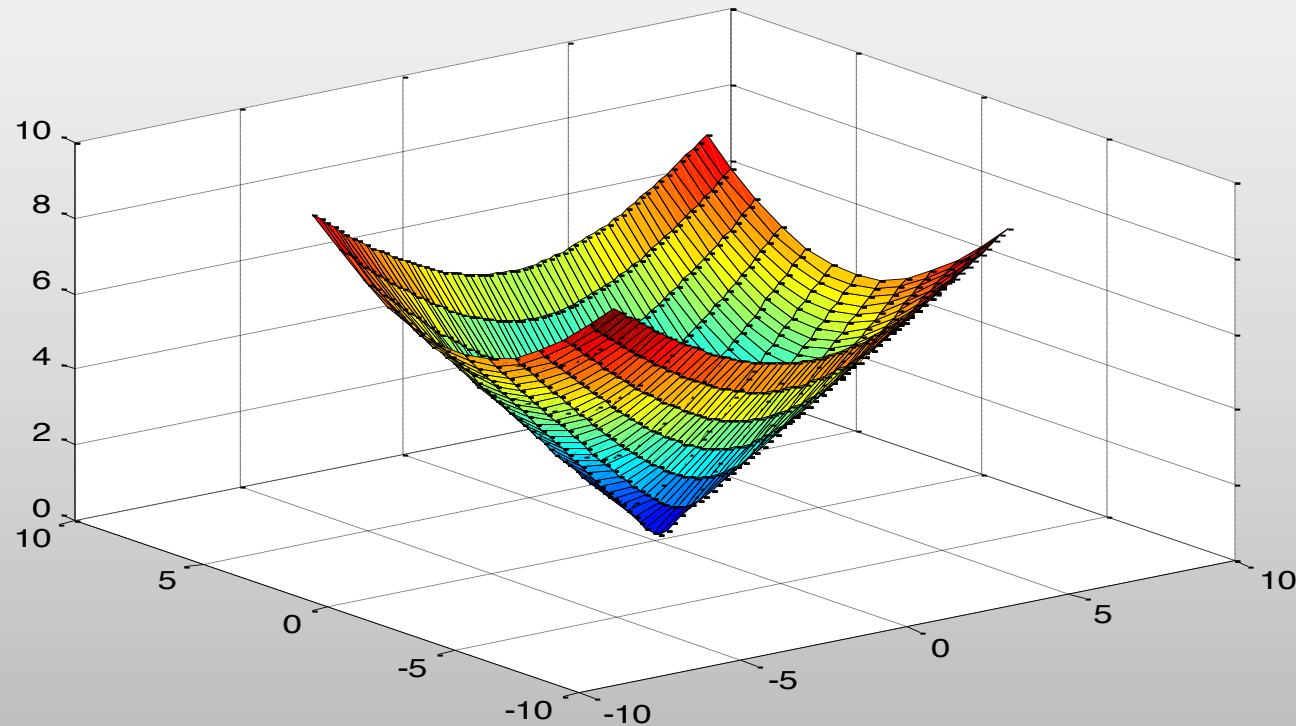
Increasing values of z



MIMICKING CONVERSION OF LEARNING TO REASONING/ACTION

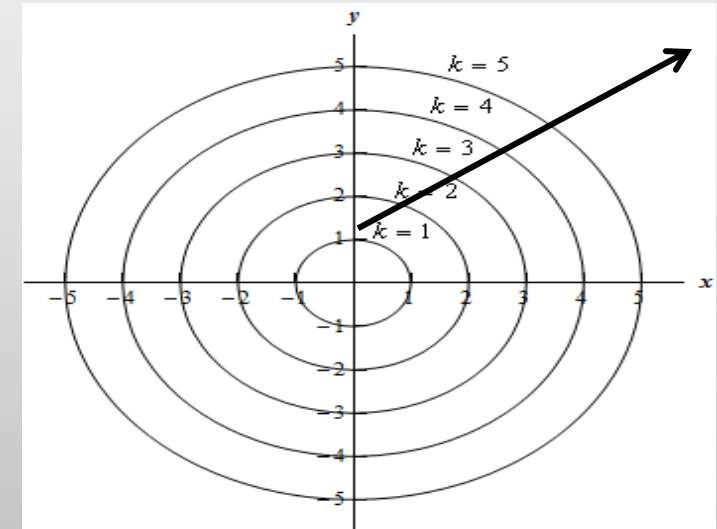
What-if analysis ?

Optimization to identify actions

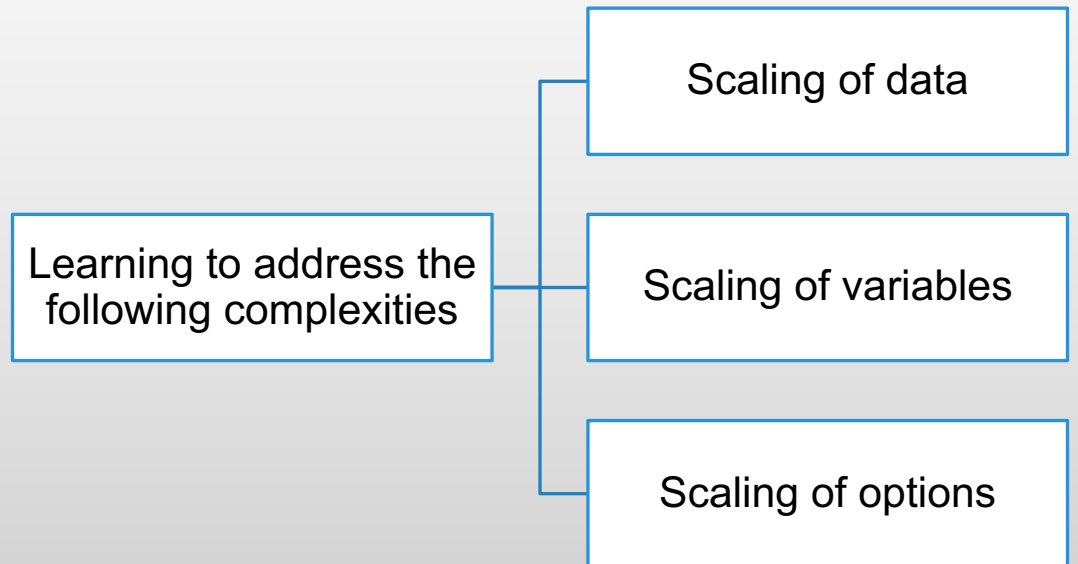
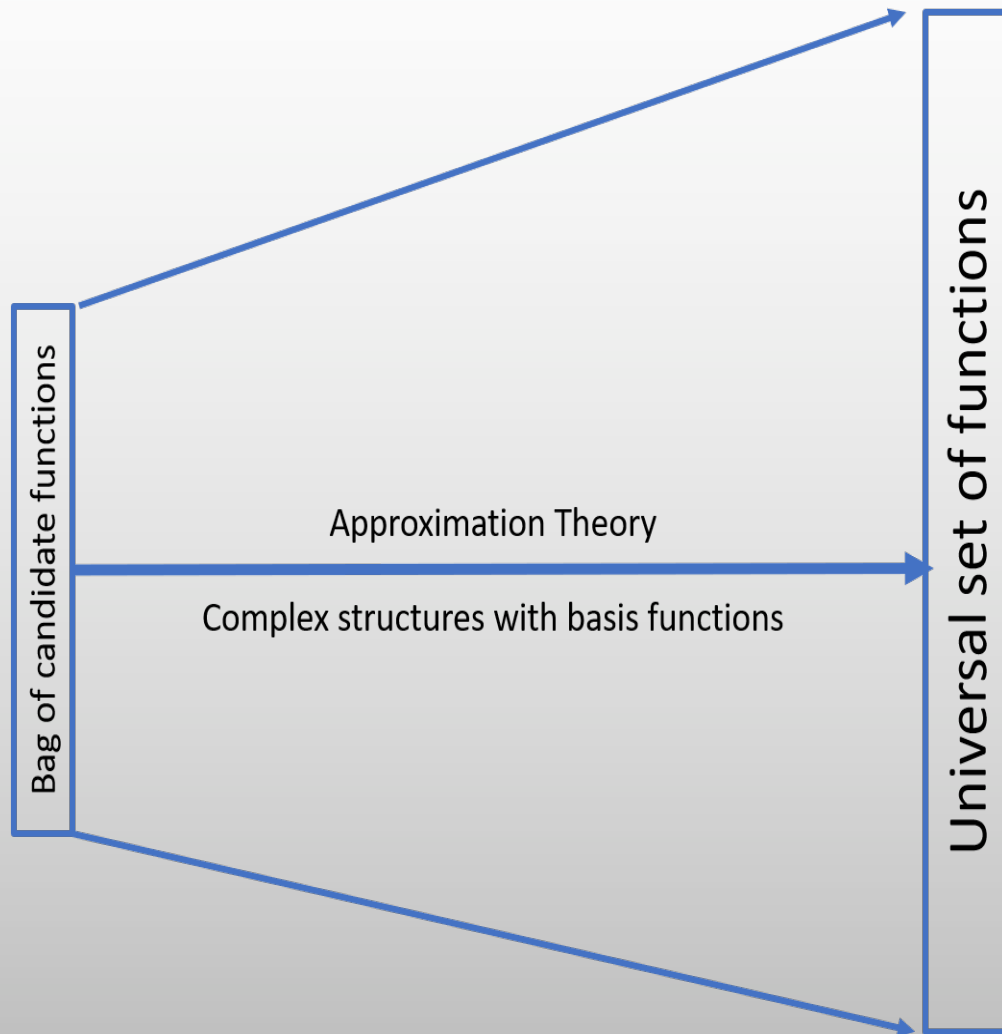


Contour plot

Increasing values of z



MACHINE LEARNING FUNDAMENTALS



Machine Learning – Classification Problems

- FRAUD DETECTION IN CREDIT CARD TRANSACTIONS
- IDENTIFYING WHETHER AN EMAIL IS A SPAM OR GENUINE
- DISTINGUISHING BETWEEN DIFFERENT PRODUCTS IN RETAIL STORE SHELF
- DETECTING DEFECTIVE AND NON-DEFECTIVE ITEMS

MACHINE LEARNING – PREDICTION PROBLEMS

- PREDICTING MATERIALS PROPERTY FOR DIFFERENT MOLECULAR STRUCTURES
- PREDICTING SCORES IN A CRICKET MATCH
- PREDICTING REACTION KINETIC PARAMETERS IN BIOCHEMICAL REACTIONS
- PREDICTING THE VALUE OF A BOARD POSITION IN A GAME OF CHESS

THE KEY IDEA BEHIND UNIVERSALITY

- ML TECHNIQUES ARE QUITE UNIVERSALLY APPLICABLE BECAUSE OF THE NOTION OF FEATURES
- PREDICTING MATERIALS PROPERTY FOR DIFFERENT MOLECULAR STRUCTURES
 - TYPES AND NUMBERS OF ATOMS
 - 1D DESCRIPTORS – STRUCTURAL FRAGMENTS, GROUPS
 - 2D DESCRIPTORS – ANGLES, GRAPH PROPERTIES
 - 3D DESCRIPTORS – VOLUME, STERIC
 - DERIVED DESCRIPTORS – FIRST PRINCIPLES CALCULATIONS

MACHINE LEARNING TECHNIQUES –TYPE - PREVALENT USE (MODULO GREY AREAS)

- K-nearest-neighbour – classification & function approximation
- K-means clustering – classification
- Linear discriminant analysis (LDA) - classification
- Logistics regression - classification
- Linear regression - function approximation
- Support vector machines - classification
- Decision trees and random - classification & function approximation

MACHINE LEARNING TECHNIQUES

- Principal component analysis - function approximation
- Quadratic discriminant analysis (QDA) – classification
- Naïve Bayes classifier - classification
- Hierarchical clustering - classification
- Neural networks – classification & function approximation

WHAT IS AI?

- MIMICKING HUMAN INTELLIGENCE (AGI – ARTIFICIAL GENERAL INTELLIGENCE)
- COULD INCLUDE MANY ASPECTS OF INTELLIGENCE
 - LEAPS OF INSIGHT, TACIT KNOWLEDGE, EMERGENCE, CONSCIOUSNESS, META-COGNITION – AWARENESS OF ONE'S OWN THOUGHT PROCESS
- KEY FEATURES
 - REPRESENTATION, REASONING AND DECISION MAKING
- VERY DIFFICULT TO REALIZE IN ITS MOST EXPANSIVE CONCEPTUALIZATION
- WHAT WE SEE NOW IS MORE APPLIED (NARROW) AI

WHAT DOES CURRENT AI INCLUDE?

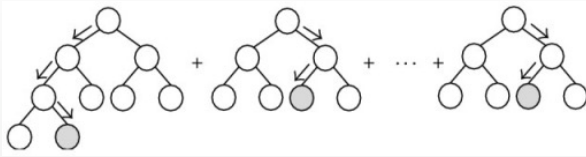
- GAMES - STUDY OF STATE SPACE SEARCH, E.G., CHESS, GO
- AUTOMATED REASONING AND THEOREM PROVING, E.G., LOGIC THEORIST
- ROBOTICS AND PLANNING – E.G., DRIVERLESS CARS
- VISION – E.G., FACIAL RECOGNITION
- NATURAL LANGUAGE UNDERSTANDING AND SEMANTIC MODELING, E.G., SIRI
- EXPERT SYSTEMS OR KNOWLEDGE-BASED SYSTEMS

WHAT DOES CURRENT AI INCLUDE?

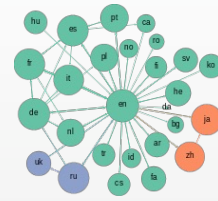
- **MACHINE LEARNING** – E.G., BAYESIAN CLASSIFIERS, DEEP NEURAL NETS
– **CURRENT APPROACH TO ACHIEVING NARROW AI**
- REINFORCEMENT LEARNING
- AUTOMATIC PROGRAMMING
- HARDWARE FOR AI
- DISTRIBUTED & SELF-ORGANIZING AI – E.G., DRONE SWARMS
- ARTIFICIAL LIFE – E.G., CELLULAR AUTOMATA, AGENT-BASED MODELING

AI/ML FRAMEWORKS FOR ENGINEERING

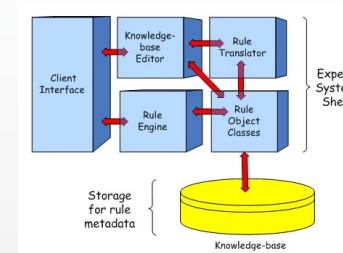
Engineers are familiar with ODE/PDE, statistical regression, and mathematical programming models. Other AI formalisms include



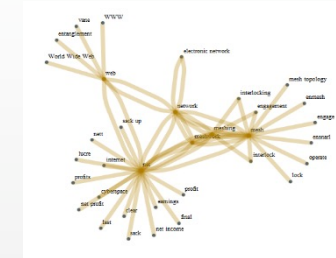
Gradient boosting
(ensemble-based methods)



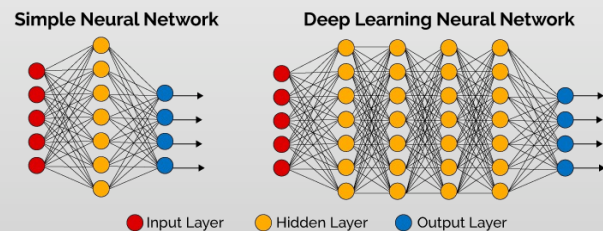
Graph-theoretical models
(causal reasoning in the identification of abnormal events, diagnosis, and risk analysis)



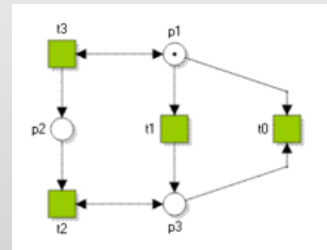
Rule-based production system models
(expert systems for automating higher-order reasoning)



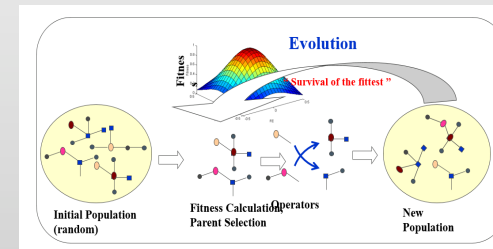
Semantic network models such as ontologies
(material discovery and design, domain-specific compilers)



Pattern recognition-based models
(neural networks, deep learning, fuzzy logic)



Petri nets
(modeling discrete event systems)



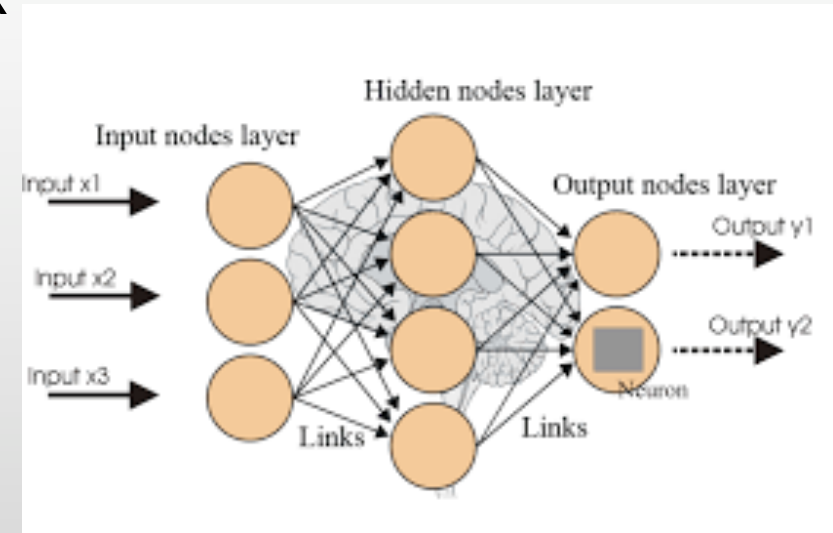
stochastic models
(genetic algorithm, simulated annealing)



Agent-based models
(simulating the behavior and decision-making choices of independent, interacting, entities endowed with complex attributes and decision-making power)

BACK-PROPAGATION NEURAL NETWORKS TO DEEP LEARNING

- NEURAL NETWORKS QUITE POPULAR IN 1980S
- WENT OUT OF FASHION DUE TO
 - NOTION THAT LEARNING NOT POSSIBLE WITHOUT PRIOR KNOWLEDGE
 - LARGE TRAINING TIMES, LOCAL MINIMA PROBLEMS
- RESURGENCE THROUGH DEEP LEARNING
 - MULTILAYERS WITH SIMPLE PROCESSING UNITS
 - REPRESENTATIONAL LEARNING
 - AUTOMATICALLY UNCOVER REPRESENTATIONS FOR HIGHER LEVEL TASKS



DEEP LEARNING – WHAT IS IT ?

- CYBENKO THEOREM

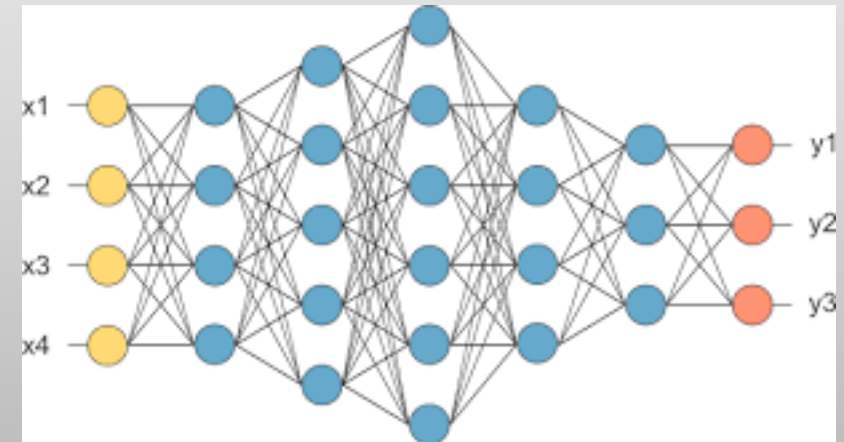
- IN SIMPLE TERMS, SINGLE LAYER NETWORKS WITH FINITE NODES ARE UNIVERSAL APPROXIMATORS UNDER SOME MILD ASSUMPTIONS

- DEEP LEARNING

- MULTIPLE LAYERS TO LEARN HIERARCHICAL FEATURES
 - FEATURES ARE LEARNT AND NOT ENGINEERED

- SUCCESSES IN RECENT YEARS

- IMAGE RECOGNITION
 - SPEECH RECOGNITION



DEEP LEARNING – WHAT IS NEW ?

- ARCHITECTURE (REPRESENTATION)
 - MULTIPLE LAYERS
 - DEEP VS BROAD
 - CONVOLUTION LAYERS, POOLING LAYERS AND FULLY CONNECTED LAYERS
- NONLINEARITY (REPRESENTATION AND LEARNING)
 - RECTIFIED LINEAR UNIT AND OTHER UNITS
- LEARNING ALGORITHM (LEARNING)
 - PRE-TRAIN LAYER BY LAYER THROUGH AUTO-ENCODING
 - FINE-TUNE ONCE ALL LAYERS ARE ASSEMBLED WITH AN OUTPUT LAYER
- IMPROVEMENTS IN OPTIMIZATION TECHNIQUES – STOCHASTIC OPTIMIZATION, GLOBAL OPTIMIZATION (LEARNING)

AI – IS IT NEW OR WHAT HAS CHANGED ?



Computing in 1985*



- Cray-2 supercomputer
- 1.9 GFLOPS
- 150 kW!
- \$40 Million! (2015 dollars)

Computing in 2015



- 3 GFLOPS
- 1 W
- \$300

Re- enter AI

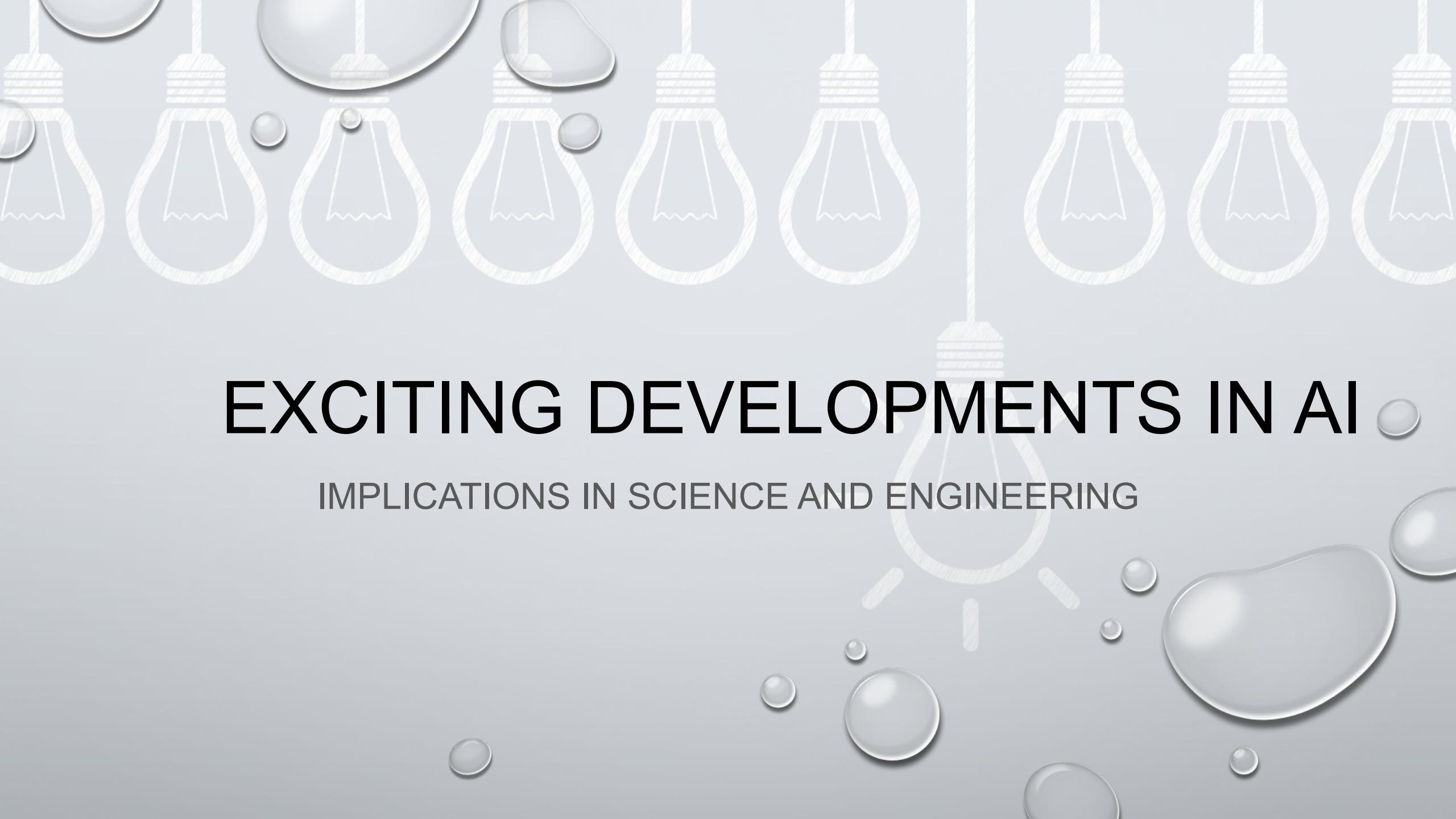


- Algorithms for deriving useful information about state of the system (engineered, social, biological)
- All-weather algorithms for all kinds of problems
- Self-learning

Industry 4.0 + DS



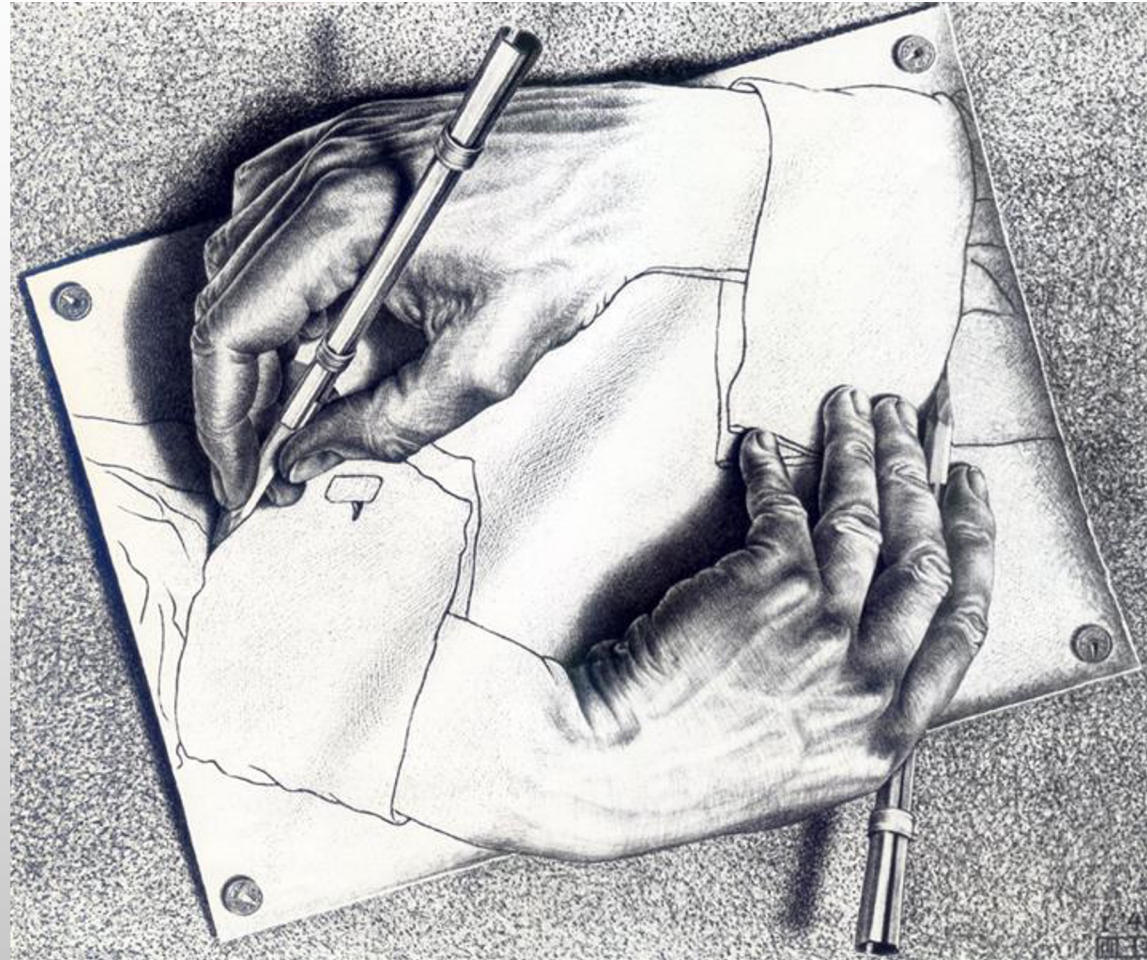
- Benchmarks
- Transfer learning
- Uncover patterns
- Power of information

The background is a light gray gradient. At the top, there is a row of lightbulbs. Some are simple white outlines, while others are filled with a textured, hand-drawn style. Below the row, there are several water droplets of various sizes, some with highlights and shadows, giving them a 3D appearance. The text is centered in the middle of the image.

EXCITING DEVELOPMENTS IN AI

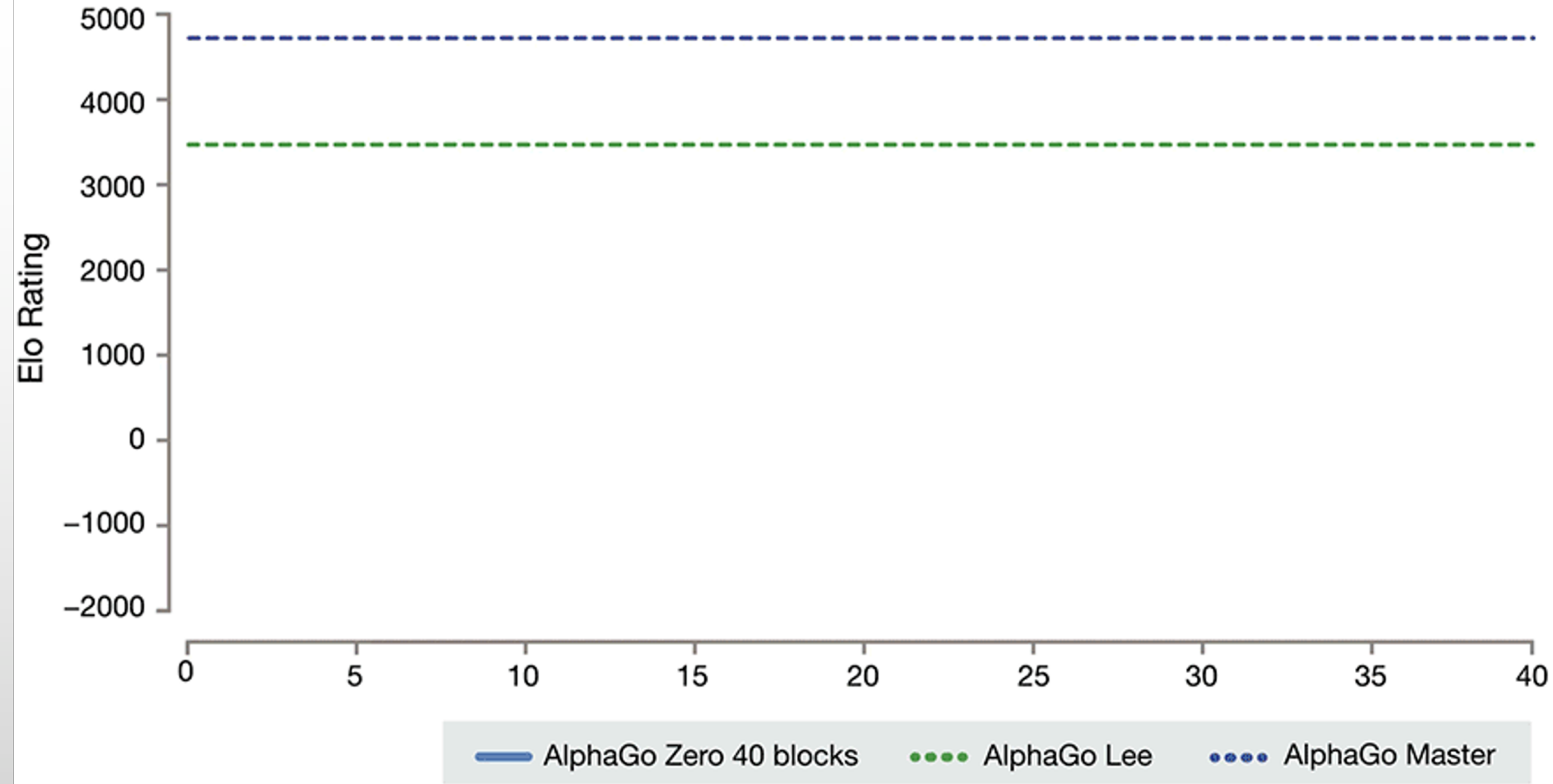
IMPLICATIONS IN SCIENCE AND ENGINEERING

SUPERHUMAN AI: HOW FAR?



AI BOTS AND GAMES

Deep Blue	- Chess (CMU, games of perfect information, 1996)
Watson	- Jeopardy! (IBM, natural language processing, 2011)
DeepMind	- Atari games (DeepMind, general AI, 2013)
AlphaGo	- Go (DeepMind, games of perfect information, 2016)
AlphaGo Zero	- Chess, Go, Shogi (DeepMind, games of perfect information, 2017) (only rules of games known)
Libratus	- Poker (CMU, games of imperfect information, 2017)
AlphaStar	- StarCraft II (DeepMind, games of imperfect information, 2019)
MuZero	- (DeepMind, uncertain environments needing planning, 2020) (without knowing rules of the game)



TRAINED WITHOUT DATASETS, ONLY RULES KNOWN - “NO LONGER CONSTRAINED BY THE LIMITS OF HUMAN KNOWLEDGE”

COUNTERINTUITIVE INSIGHT ?

SOME OF ITS MOVES, SUCH AS MOVING THE KING TO THE CENTRE OF THE BOARD, GO AGAINST SHOGI THEORY AND - FROM A HUMAN PERSPECTIVE - SEEM TO PUT ALPHAZERO IN A PERILOUS POSITION. BUT INCREDIBLY IT REMAINS IN CONTROL OF THE BOARD. ITS UNIQUE PLAYING STYLE SHOWS US THAT THERE ARE NEW POSSIBILITIES FOR THE GAME.

- *YOSHIHARU HABU, 9-DAN PROFESSIONAL, ONLY PLAYER IN HISTORY TO HOLD ALL SEVEN MAJOR SHOGI TITLE*

MOST UNNERVING WAS THAT ALPHAZERO SEEMED TO EXPRESS INSIGHT. IT PLAYED LIKE NO COMPUTER EVER HAS, INTUITIVELY AND BEAUTIFULLY, WITH A ROMANTIC, ATTACKING STYLE....IN GAME 10, ALPHAZERO RETREATED ITS QUEEN BACK INTO THE CORNER OF THE BOARD ON ITS OWN SIDE, FAR FROM STOCKFISH'S KING, NOT NORMALLY WHERE AN ATTACKING QUEEN SHOULD BE PLACED...

YET THIS PECULIAR RETREAT WAS VENOMOUS,...

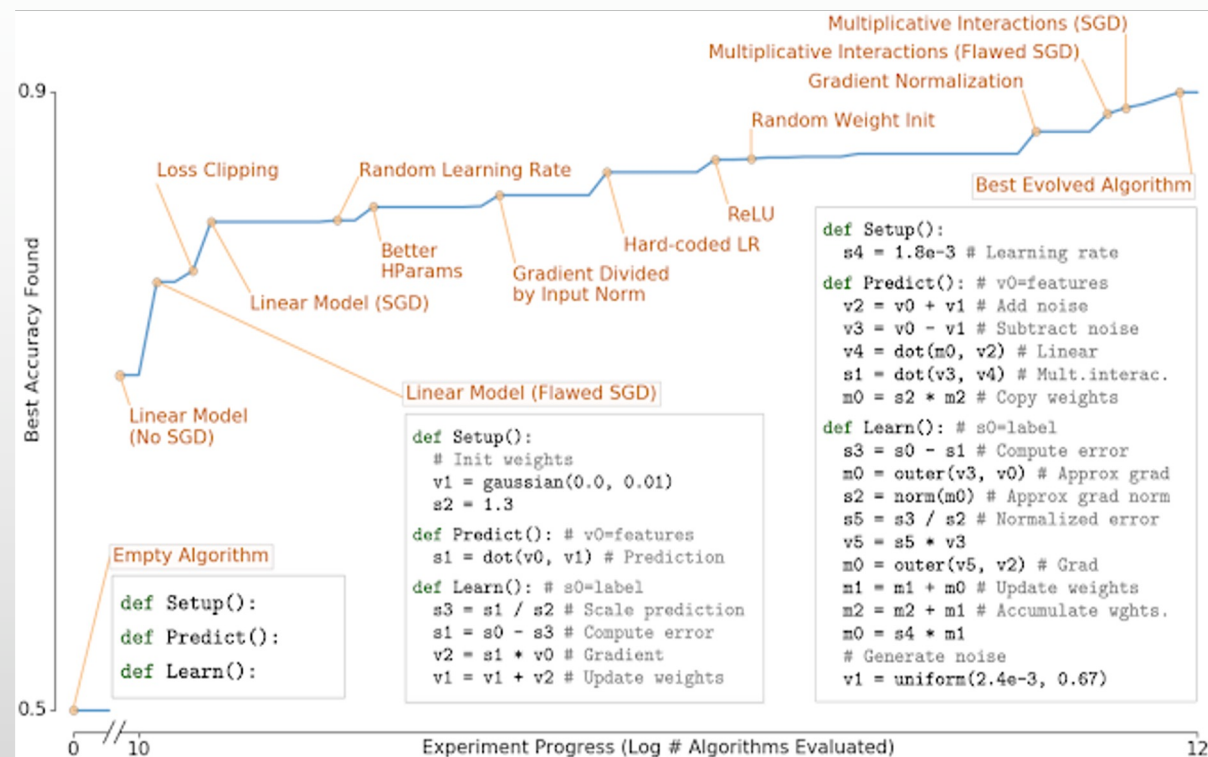
GRANDMASTERS HAD NEVER SEEN ANYTHING LIKE THIS ...

IT WAS HUMANKIND'S FIRST GLIMPSE OF AN AWESOME NEW KIND OF INTELLIGENCE

- *STROGATZ, NYT, DEC 2018*

AI DESIGNING AI

- DISCOVER MACHINE LEARNING ALGORITHMS FROM SCRATCH
- USING EVOLUTIONARY SEARCH
- LIBERATED FROM THE BIASES OF HUMANS



AI MATHEMATICIAN

- RAMANUJAN MACHINE : CONJECTURING CONTINUED FRACTION FORMULA, KNOWN AND UNKNOWN, FOR FUNDAMENTAL CONSTANTS, THUS UNVEILING UNDERLYING MATHEMATICAL STRUCTURE (RAAYONI G ET AL., NATURE, 2021)
- SYMBOLIC MATH : SOLUTION TO 5000 PROBLEMS (LAMPLE G ET AL., ICLR, 2020)
- HOLIST : GOOGLE'S AI MATHEMATICIAN PROVES 1200 THEOREMS (BANSAL K ET AL., ICML, 2019)
- THEOREM PROOFS: FOUR COLOR PROBLEM, KEPLER CONJECTURE, OPTIMAL SOLUTION TO RUBIKS CUBE

AI RESEARCHER

- AUTOMATING SCIENCE : CARRYING SCIENTIFIC RESEARCH WITH MINIMAL HUMAN INTERVENTION
- ROBOT SCIENTIST, ADAM : EFFICIENTLY IDENTIFY WHICH GENES ENCODE PARTICULAR ENZYMATIC ACTIVITIES REQUIRED FOR YEAST GROWTH. (KING RD ET AL., SCIENCE, 2009)
- ACTIVE MACHINE LEARNING (QUERY FOR DATA NOT READILY AVAILABLE) TECHNIQUES (NAIK ET AL., ELIFE, 2016)
 - ML WITH ROBOTICS AND AUTOMATED MICROSCOPY WITHOUT PRIOR KNOWLEDGE LEARNED THE EFFECTS OF 48 CHEMICAL COMPOUNDS ON THE SUBCELLULAR LOCALIZATION OF 48 PROTEINS WHILE PERFORMING ONLY 29% OF ALL POSSIBLE EXPERIMENTS
- IRIS.AI : SPEED-READERS TO HANDLE LARGE VOLUME OF PUBLICATIONS BY CREATING DOCUMENT FINGERPRINTS
- EURETOS :RELIES ON ONTOLOGIES TO GENERATE HYPOTHESIS. USED TO IDENTIFY GENE-EXPRESSION CHANGES IN SPINOCEREBELLAR ATAXIA TYPE 3 (TONEEN L ET AL., 018)
- MASTER PROGRAM LAUNCHED AT CMU IN AUTOMATED SCIENCE (2018)

OTHER APPLICATIONS

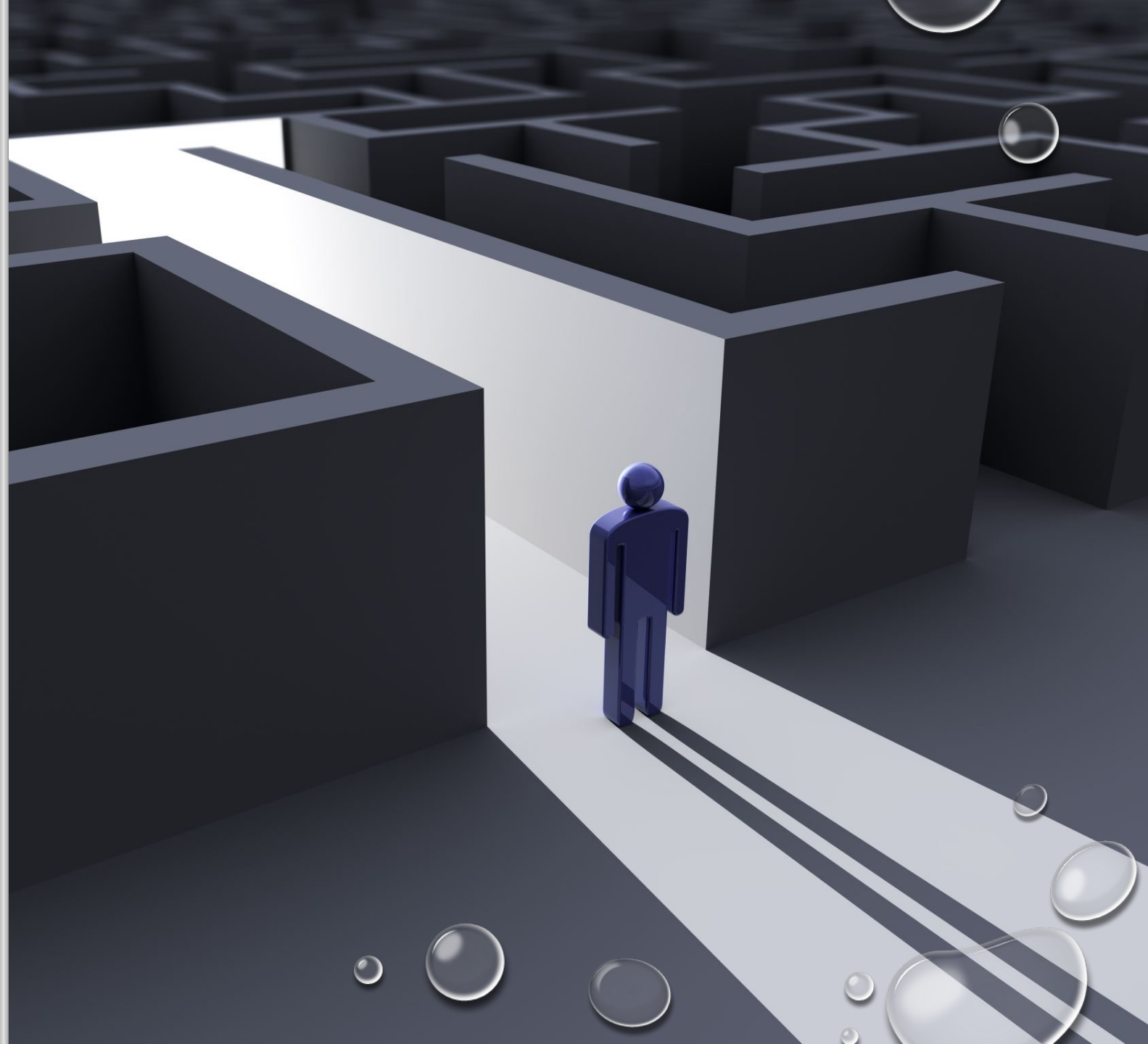
- ALPHAFOLD (2018), ALPHAFOLD 2 (2021) - PROTEIN FOLDING (ACCURATE PREDICTION OF STRUCTURE OF SARS-COV-2 SPIKE PROTEIN)
- PREDICTION OF EXUDATIVE AGE-RELATED MACULAR DEGENERATION (2020)
- DETECTION OF ACUTE NEUROLOGICAL EVENTS, SUCH AS STROKE FROM CRANIAL IMAGES (2018)

CRITIQUE OF AI

- TASK ORIENTED AI
- LARGE TRAINING DATASETS
- LARGE COMPUTATIONAL COSTS
- WELL-DEFINED DEMONSTRATION EXAMPLES
 - HANDLING UNCERTAINTY IS DEFINITELY A CHALLENGE
- WHAT DOES IT TAKE TO SIMULATE 7 BILLION HUMAN BRAINS? (NOT SUSTAINABLE USING CURRENT ARCHITECTURE)

WHERE ARE WE GOING ?

- FROM SUPERVISED TO UNSUPERVISED LEARNING
- FROM “NARROW” AI TO GENERAL AI
- FROM BRUTE FORCE SEARCH TO SOPHISTICATED LEARNING



Future Directions

- Similar to AlphaZero coming up with moves that are counterintuitive, yet, super intelligent, it is time algorithms evolve their own architectures that are super-efficient
- Game theory could help evolution of neural architectures
- The future super-efficient neural architectures are “speculated” to have an underlying simplicial structure (evidence of this has been found in topological structure of connectivity in brains)